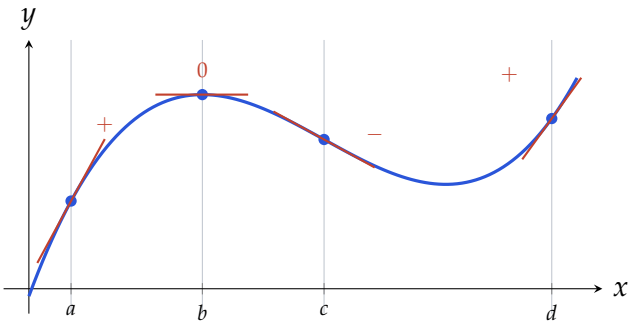
Lesson 1-7 • Instantaneous Rates of Change

Notes (filled in)

Objective: draw tangent lines, read the sign of the derivative, and recognize where a graph is not differentiable.

Rates on a graph



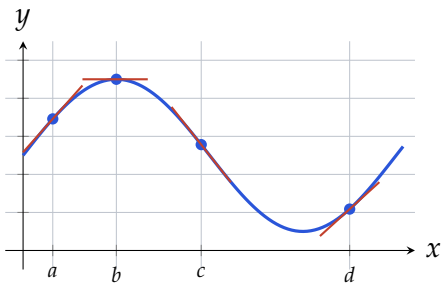
The instantaneous rate of change at a point is the slope of the tangent line : the line that just touches the graph at that point.

Synonyms

Instantaneous rate of change = slope of the tangent line = derivative = $f'(c)$ = $\frac{dy}{dx}$

Example 1

Sketch the tangent line at each point and decide whether the derivative is positive, negative or zero.



$g'(a)$ > 0

$g'(b)$ $= 0$

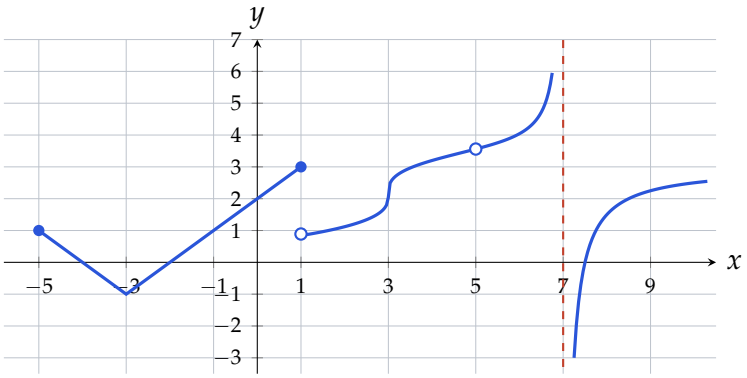
$g'(c)$ < 0

$g'(d)$ > 0

Differentiability

A function is **not differentiable** (f' does not exist) at:

1) discontinuities: holes, jumps, vertical asymptotes, endpoints
2) sharp corners (cusps)
3) vertical tangent lines

Example 2	Find the x-coordinates where h is not differentiable, and say why.  $x = -5$: endpoint. $x = -3$: sharp corner. $x = 1$: jump (discontinuous). $x = 3$: vertical tangent. $x = 5$: hole (discontinuous). $x = 7$: vertical asymptote.
Summary	The derivative $f'(c)$ is the slope of the tangent line at $x = c$: positive where the graph rises, negative where it falls, zero at a peak or valley. It does not exist at discontinuities, sharp corners and vertical tangents.